

Logic gates & Boolean algebra

Binary digits

Bi $\rightarrow$ Two

0	1	} Complete Philosophy of Binary digit.
- low value	high value	
- off	On	
- false	True.	

The computer uses Binary digit.

So in boolean algebra. all mathematical operation are the same except $1 + 1$.

Addition

$$0 + 0 = 0$$

$$0 + 1 = 1$$

$$1 + 0 = 1$$

$$1 + 1 = 1$$

Multiplication

$$0 \cdot 0 = 0$$

$$0 \cdot 1 = 0$$

$$1 \cdot 0 = 0$$

$$1 \cdot 1 = 1$$

Understanding logic gate, the knowledge of boolean algebra is needed.

Concepts of Boolean algebra.

Basic Concepts.

Addition of Binary Digits.

$$A=0 \quad A=1$$

$$\boxed{A+1=1}$$

$$0+1=1$$

$$1+1=1$$

$$\boxed{A+0=A}$$

$$0+0=0$$

$$1+0=1$$

Multiplication of Binary Digits

$$\boxed{A \cdot 1 = A}$$

$$\boxed{A \cdot 1 = A}$$

$$0 \cdot 1 = 0$$

$$1 \cdot 1 = 1$$

$$\boxed{A \cdot 0 = 0}$$

$$0 \cdot 0 = 0$$

$$1 \cdot 0 = 0$$

Inverse Concepts.

Low value high value

$$0 \rightarrow 1$$

$$A \rightarrow \bar{A}$$

$$\boxed{A + \bar{A} = 1}$$

$$A=0 \quad \bar{A}=1$$

$$0+1=1$$

$$\boxed{\bar{\bar{A}} = A}$$

$$A=1 \quad \bar{A}=0$$

$$\bar{A}=1$$

$$\boxed{A \cdot \bar{A} = 0}$$

$$0 \cdot 1 = 0$$

Advanced concept

$$\overline{(A+B)} = \bar{A} \cdot \bar{B}$$

Proof:

$$\text{LHS: } \overline{A+B}$$

$$A=1 \quad B=0$$

$$(A+B)$$

$$\overline{(A+B)}$$

$$A+B=1$$

$$1+0=1$$

$$\overline{(A+B)} = 0 \Rightarrow \text{eq 1}$$

$$\therefore \overline{A+B} = \bar{A} \cdot \bar{B}$$

$$\text{RHS: } \bar{A} \cdot \bar{B}$$

$$A=1, B=0$$

$$\bar{A}=0, \bar{B}=1$$

$$\bar{A} \cdot \bar{B} = 0 \cdot 1$$

$$= 0 \dots \text{eq 2}$$

$$\therefore \text{eq 1} = \text{eq 2.}$$

$$\text{LHS} = \text{RHS}$$

$$\therefore \overline{A+B} = \bar{A} \cdot \bar{B} \quad \text{Similarly } \overline{A \cdot B} = \bar{A} + \bar{B}$$

Concept

Basic concepts of logic gates

What is the basic purpose of a gate.

open gate will allow passage.

A closed gate will not allow passage.

Similarly, a logic gate that is

A closed gate will allow electrical signal to pass through.

closed will not allow electrical signals to pass through

Logic gates are used to allow or block electrical (digital) signals to enter into the circuit.

Logic gates are used to on and off any digital circuits.

What are logic gates?

Two inputs $\rightarrow$ One output.

"They are digital devices which perform boolean function by taking two inputs and giving one output"



Allow electrical signals



Don't allow electrical signals

When we give two inputs to the logic gates, it performs a Boolean Function and returns an output.

So logic gates work on the principles of boolean algebra

Logic gates are made up of diodes and transistors.

So with the help of boolean algebra, logic gates allow or block signals from entering into the circuit.

Types of Logic Gates.

There are 5 Universal Logic gates

OR

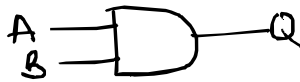


$$Q = A + B$$

A	B	Q
0	0	0
0	1	1
1	0	1
1	1	1

OR gate always sum up the value.

AND



$$Q = A \cdot B$$

A	B	Q
0	0	0
0	1	0
1	0	0
1	1	1

and gate always multiple the values

NOT



$$Y = \bar{A}$$

A	Y
0	1
1	0

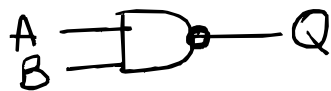
(OR + NOT)
NOR



$$Q = \overline{A + B}$$

A	B	Q
0	0	1
0	1	0
1	0	0
1	1	0

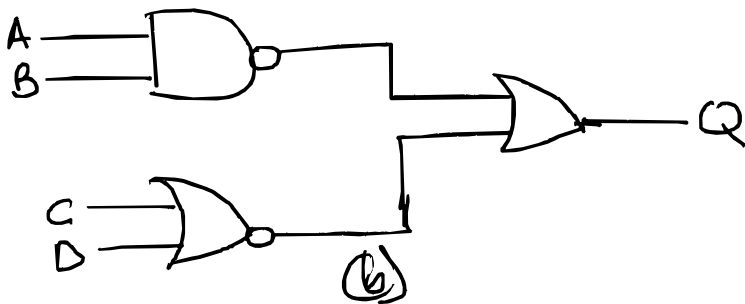
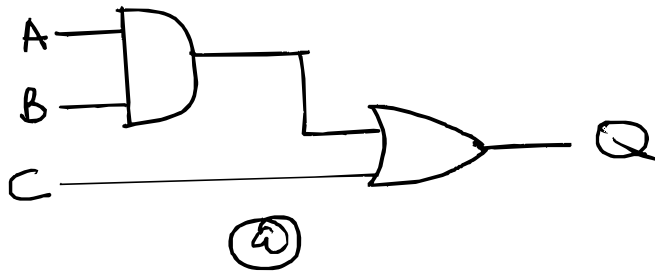
(AND + NOT)
NAND



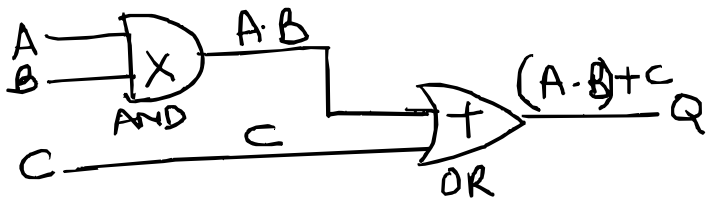
$$Q = \overline{A \cdot B}$$

A	B	Q
0	0	1
0	1	1
1	0	1
1	1	0

Writing functions for logic gates.

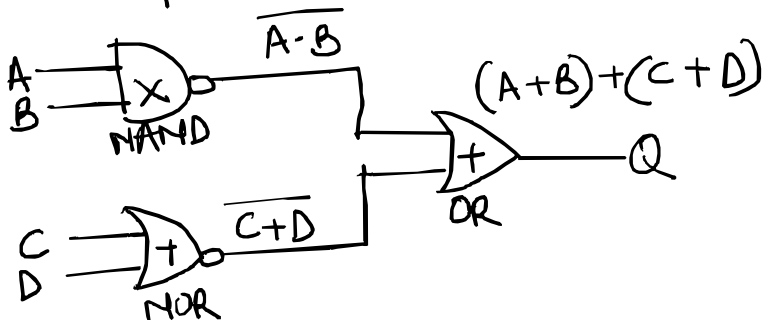


For the above (a) and (b) logic gate based system. The function equations are written as follows



$$Q = (A \cdot B) + C$$

For system b



$$Q = \overline{A \cdot B} + \overline{C + D}$$

Example question.

What will be the value of A and B for the boolean expression $\overline{A+B} \cdot (\overline{A} \cdot \overline{B}) = 1$?

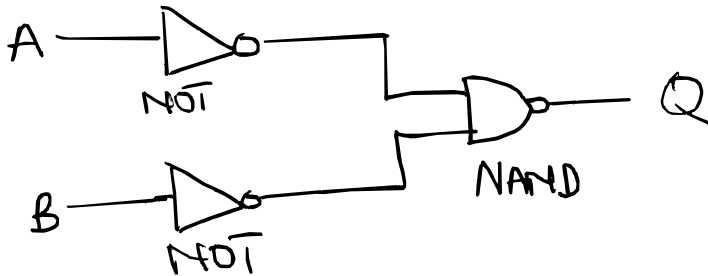
$$A = 0 \quad B = 0$$

$$\begin{aligned} \text{L.H.S: } & \overline{A+B} \cdot (\overline{A} \cdot \overline{B}) \\ & \overline{0+0} \cdot (\overline{0} \cdot \overline{0}) \\ & = (\overline{0}) \cdot 1 \cdot 1 \end{aligned}$$

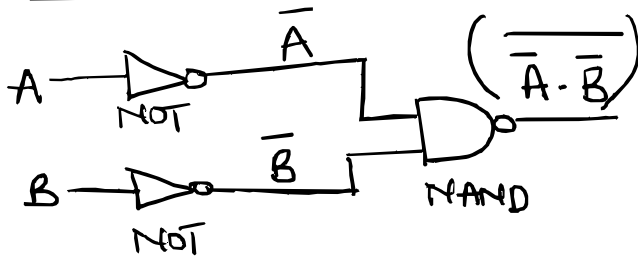
$\therefore$ The value of A and B to make the boolean expression $\overline{A+B} \cdot (\overline{A} \cdot \overline{B}) = 1$ is

$$\boxed{A = 0 \text{ and } B = 0}$$

This logic gate is equivalent to



Answer



$$\overline{\bar{A} \cdot \bar{B}} = \overline{\bar{A} + \bar{B}} = A + B$$

$\therefore$ The digital circuit is equivalent to OR gate